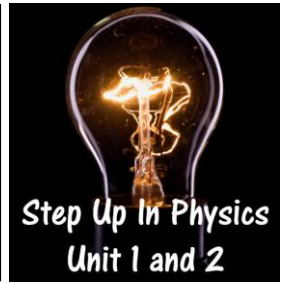
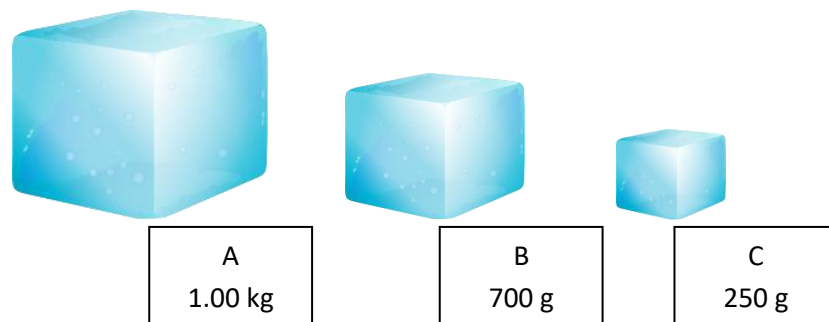


Latent Heat

Problems Worksheet



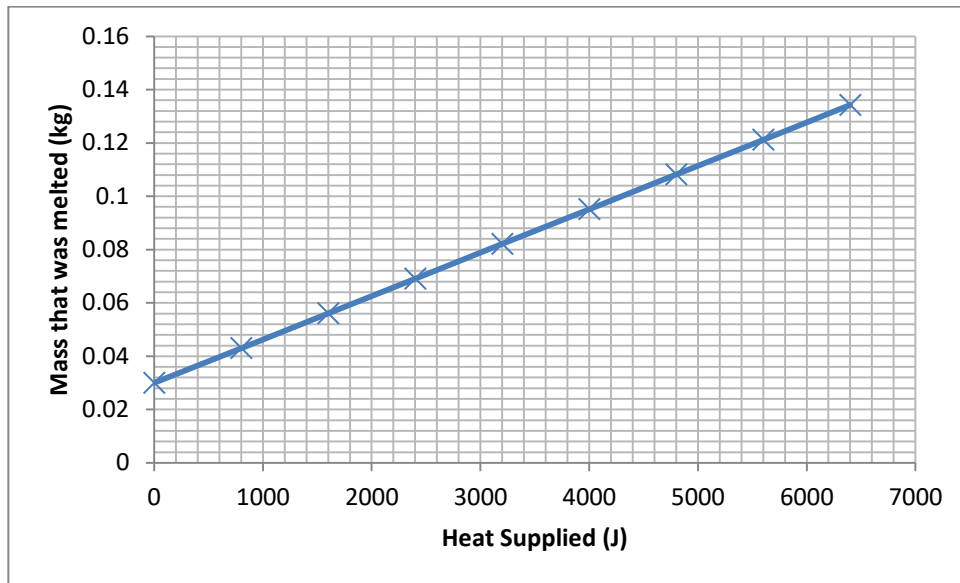
1. It takes 8.00 MJ to completely boil a 4.20 kg sample of liquid sitting at its boiling point. Assuming no loss of energy to the surroundings, calculate the latent heat of vaporisation of the liquid.
2. The units for latent heat are $J\ kg^{-1}$ while the units for specific heat capacity are $J\ kg^{-1}\ K^{-1}$. Explain the difference between the two units.
3. The three ice cubes shown below have different masses.



- a. Which ice cube (A, B, or C) has the greatest latent heat of fusion? Justify your answer.
- b. When left outside on a 30°C day, explain which ice cube would take the longest to melt. Refer to relevant physical formula and theory in your explanation.

4. When the human body gets too hot it perspires (sweats) in an attempt to cool down. Explain how perspiring can help to cool a human body.
5. On a very windy, 35°C day the pool in a backyard has a temperature of 24°C . Despite the water being at a lower temperature than the air, when the children get out of the pool they feel colder outside the water than in. Explain why this occurs.
6. A saucepan of temperate (i.e. room temperature) water is placed on a stove. The stove is on and transfers thermal energy into the water until no more water remains in the saucepan. Describe the energy transformations that occurred during this time.

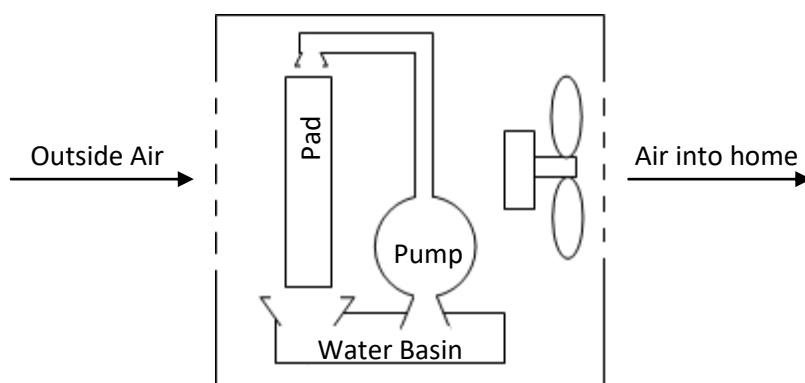
7. The graph below shows the results for an experiment designed to determine the latent heat of fusion of a substance. Two students used the heat produced by a Bunsen burner (according to the manufacturing specifications) and the mass of that substance had melted inside a beaker. The students who took the data waited until some of the solid substance had clearly melted before taking measurements to ensure the substance was at its melting point.



- a. Using the gradient of the graph, calculate the latent heat of fusion.
- b. Describe a major source of error likely to have affected the student's determination of the latent heat of fusion. In your description, clearly state whether the actual latent heat of fusion would be higher or lower than calculated.

8. The latent heat of vaporisation is orders of magnitude higher than the latent heat of fusion. Explain why this is.

9. An evaporative air conditioning system uses a pump to spray water over pads. A ventilator fan pulls outside air through the pads and into the home.



- a. Will this system cool the house or warm the house?
- b. Explain how the air temperature in the home is changed by the system.
- c. Evaporative air conditioners lose effectiveness on humid days. Explain the loss of effectiveness.

10. A quantity of heat was used to melt a block of ice at 0°C . This same quantity of heat was only able to melt 20.0% of an equal mass of an unknown substance. Calculate the latent heat of fusion of the unknown substance.

Find the heat required to melt the ice. As the mass is unknown, the energy will be in terms of mass.

$$Q = mL_f = m * 3.34 * 10^5$$

This quantity of heat was used to melt 20% of another mass m .

$$Q = mL_f \quad \therefore \quad L_f = \frac{Q}{m} = \frac{m * 3.34 * 10^5}{0.20 * m} = 1.67 * 10^6 \text{ J kg}^{-1}$$